

SPECIAL ISSUE ARTICLE

Fostering STEAM through challenge-based learning, robotics, and physical devices: A systematic mapping literature review

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Abstract

Nowadays, companies are demanding better-prepared professionals to succeed in a digital society, and the acquisition of Science, Technology, Engineering, Arts, and Mathematics (STEAM)-related competencies is a key issue. One of the main problems in this sense is how to integrate STEAM into the current educational curricula. This is not something related to a subject or educational trend but rather to new methodological approaches that can engage students. In this sense, such active methodologies that apply mechatronics and robotics could be an interesting path to pursue. Given this context, the first necessary task in evaluating the potential of this approach is to understand the landscape of the application of robotics and mechatronics in STEAM Education and how active methodologies are applied in this sense. To carry out this analysis in a systematic and replicable manner, it is necessary to follow a methodology. In this case, the researchers employ a systematic mapping review. This paper presents this process and its main findings. Fifty-four studies have been selected out of 242 total studies analyzed. From these, beyond obtaining a clear vision of the STEAM landscape regarding project topics, we can also conclude that robotics and physical devices have been applied successfully with collaborative methodologies in STEAM Education. Regarding conclusions, this paper shows that robotics and mechatronics applied with active methodologies is to be a good means to engage students in STEAM disciplines and thus aid the acquisition of what is commonly known as “21st-century skills.”

KEYWORDS

challenge-based learning, mechatronics, physical devices, problem-based learning, project-based learning, robotics, STEAM, STEM

1 | INTRODUCTION

In our current changing society, educational systems should be flexible enough to provide students with the

competencies that the labor market demands of them. The digital society requires persons able to find the best solutions in a continuously evolving context. To do this, these candidates need to know their environment very

well, as well as the applicable methodologies, and have sufficient knowledge to deal with very different and changeable devices, apps and tools; and they also need to develop the skills and acquire competencies in applying that knowledge properly [47,71].

In this context, educational institutions should devote lots of effort to providing their students with competencies such as computational thinking, problem-solving, programming, and so forth [3,9,18]. This is the way to guarantee students' employability and turn out well-qualified professionals. Such competencies can be acquired through STEAM (Science, Technology, Engineering, Arts, and Mathematics) [54] Education, but the problem is how to integrate these disciplines into current educational curricula. One possibility is to add some subjects to these curricula, but this solution is not sufficient because STEAM will then appear as something isolated from other disciplines, where it could be used as a solution. From this point of view, the integration does not require new content, but the application of new learning methodologies and tools [47]. Active and collaborative methodologies facilitate addressing this problem [46]. They are very popular and have shown to be useful in very different contexts. Some examples could be project-based learning (PrBL) [95], problem-based learning (PBL) [19] and challenge-based learning (ChBL) [60]. Regarding the tools to be employed when applying these methodologies, students should achieve better results when they use tools that engage them (because they are popular, innovative or they feel comfortable using them). In this way, it is easier to achieve learning objectives. In addition, an interesting possibility is the use of tangible tools to engage and improve students' performance [33].

These concepts have been examined in some studies but to evaluate the potential of such approaches, it is necessary to explore the current state of the art, that is, the current landscape of STEAM Education, physical devices and robotics (PD&R), and the application of collaborative methodologies. From a preliminary examination of the literature, it was possible to conclude that there is no clear research or predominant approach describing either the effects or the best practices to apply PD&R with active methodologies in the context of STEAM Education. What the authors found were different initiatives, theoretical studies, and practical experiences but not a clear analysis of the current background. Given this situation, it is necessary to carry out a literature review using a systematic approach. In this case, as what the authors want to achieve is a common understanding of the existing studies and effects of using PD&R in STEAM Education, they decided to carry out a systematic mapping (SMP). An SMP aims to provide an overview of a research and identify the quantity and type

of research and results available within it. The SMP provides a structured report of the type of research and results that have been published by categorizing them, often giving a visual summary, the map, of its results. It often requires less effort than other methodologies, while providing a more coarse-grained overview [88,89].

It should be noted that this study has been carried out in the context of the RoboSTEAM project. It is an Erasmus+ Strategic Partnership project that involves eight partners from four different countries (four schools and four universities) [27,29]. It aims to define a methodology and a set of tools that help learners to develop computational thinking by using/programming PD&R in pre-university education stages. The project will also improve teacher education, providing them with a framework for easy STEAM integration in different educational contexts, by providing guidelines for good practices and lessons learned, adapted to those contexts. All these products will be tested in different countries and cross-validated in different educational institutions [40].

This paper presents the mapping process, its results, and the main findings obtained from them. The rest of the paper is structured as follows: Section 2 describes the SMP process, Section 3 presents the main results, and Section 4 a discussion about them; finally, the work closes with some conclusions.

2 | THE SYSTEMATIC MAPPING PROCESS

We are developing an SMP with a software engineering strategy, so, the first step is to define the main aim of the review and the second to develop it. The main objective of this review is to collect and analyze the existing studies related to the application of PD&R in education, taking into account three topics: The applied methodology, the type of devices employed, and the contexts of the students involved (educational level, age, country, etc.). Once the objective has been defined, it is necessary to complete the next two phases, planning and conducting [89]. In these, we will define a set of mapping questions (MQs) that will help to answer the research questions (RQs), that facilitate the achievement of the mapping goal (the relationship between the different questions—the objective is shown in Figure 6 in Section 4).

2.1 | Planning stage

The first stage of the process is planning the review. It includes several steps (see Figure 1): (a) Define the RQs, that is, what the review should answer; (b) define the

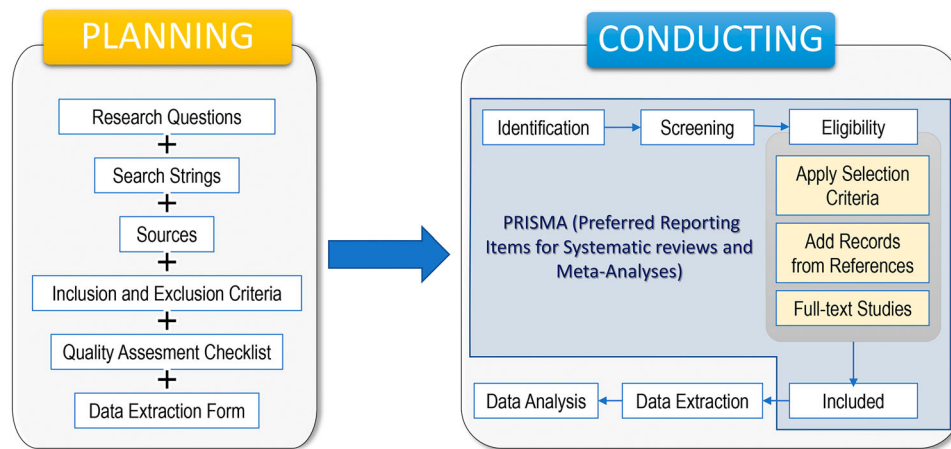


FIGURE 1 Planning and conducting main steps

search strings, including the most relevant terms related to the mapping; (c) describe the sources where the SMP will look for studies; (d) establish the selection criteria, that is, define considerations regarding excluding a study from the review; (e) define some guidelines of quality to evaluate the studies selected; and (f) define some MQs to be applied to all papers to facilitate data extraction.

2.1.1 | The research questions

To undertake the review on the basis of the main objective defined in the introduction, the following RQs were formulated:

- RQ1. What are the main approaches used in STEAM Education which apply robotics and active methodologies?
- RQ2. Which are the potential benefits over the years of changing the methods of conducting STEAM Education approaches that apply PD&R and active methodologies?
- RQ3. Which are the emerging tendencies in the application of PD&R and active methodologies in STEAM Education?

2.1.2 | The search string

Following the definition of the RQs it is necessary to establish the search string. It should include all the relevant terms to be used in the search process to answer the RQs. In this case, the terms selected were:

- “Robotics” as a word or in combination with others such as “physical devices” or “mechatronics.”

- “Education” as the context where we would like to know how the above tools are applied.
- “STEAM” or “STEM” also as part of the context for the search, using both terms because the inclusion of Arts in the acronym is newer than the other one.
- “Challenge Based Learning,” “Problem Based Learning,” and “Project Based Learning” as the possible methodologies. These terms are also used in the search using acronyms such as “PBL” and using hyphen such as in “Problem-based Learning.”
- “Computational Thinking” as one of the competencies to be developed in the RoboSTEAM project.

With these terms and employing combinations of Boolean operators (AND and OR), we defined a search string that may vary depending on the source. For instance, for ACM we need several search strings, whereas, in IEEE, one is enough. The generic search string used is the following:

((“robotics” OR “mechatronics” OR “physical devices”) AND Education AND (“STEAM” OR “STEM”) AND (“Computational Thinking” OR “Challenge Based Learning” OR “CBL” OR “Problem-based Learning” OR “Project-based Learning” OR “Problem Based Learning” OR “Project Based Learning” OR “PBL”))

2.1.3 | Source selection

The next choice to be made is to select where to look for studies. This can determine the scope of the mapping. In this case, we consider five of the most popular and complete libraries, taking into account the topics of this study:

- Springer Link (<http://link.springer.com>).

- IEEE Digital Library (<https://ieeexplore.ieee.org/>).
- SCOPUS (<https://www.scopus.com/>).
- ACM Digital Library (<https://dl.acm.org/advsearch.cfm>).
- Web of Science (www.webofknowledge.com).

The reason for applying these sources is that they are very reliable, include a very high number of journals and conferences and are the most valued ones in an information and communications technology (ICT) context.

2.1.4 | Definition of the inclusion and exclusion criteria

After establishing the sources where the search is to be carried out, it is necessary to define some criteria to allow deciding which studies can be discarded, based on their metadata, title, and abstract. Only the included papers will be used to continue the review process. For instance, we can decide that those papers written before 2000 are to be excluded, or those papers that do not include STEAM as any of their keywords. For this SMP, the inclusion criteria are:

- IC1. The work is written in English.
- IC2. The work was published after the year 2000.
- IC3. The paper was published in a peer-review journal or conference.
- IC4. The work includes one or more of the terms related to the topics of the RQs.
- IC5. The work develops an implementation initiative.

The exclusion criteria are:

- EC1. Studies for which the paper is not accessible.
- EC2. Studies that are not papers.
- EC3. Studies not published in a peer-review journal or conference.
- EC4. Studies that deal with the topics of this study and include the search terms, but do not answer the RQs.
- EC5. Studies that in discussing some of the research issues, just redefine general concepts.

2.1.5 | Checklist for quality assessment

The next step of the process, once those papers that fulfill the inclusion criteria have been included, is to check the studies' quality and retain only those with acceptable quality. To do so, we have defined a checklist of 11 questions. All the questions, except the last one, can be answered with "YES", "PARTIALLY" or "NO" and each answer has a score assigned (1, 0.5, or 0, respectively).

Regarding the last question, the most important for this SMP, it was graded from 0 to 10. The checklist items were:

- QC1. Is the work based on research or is it merely a report based on an expert opinion?
- QC2. Are the aims of the work properly stated?
- QC3. Does the work properly describe the research context?
- QC7. Is information collected in a proper way to address the research issue?
- QC4. Does the research design of the work allow addressing the research goals?
- QC5. Was the sample selection done properly according to the research aims?
- QC6. Is there a control group and an experimental group for comparing results?
- QC8. Are findings properly described and discussed?
- QC9. Does the work have an application for research or practice?
- QC10. Is the analysis of information sufficiently rigorous and supported by the literature?
- QC11. Does the paper describe the application of PD&R in education?

2.1.6 | Data extraction

The last step of the planning phase is the definition of how to extract the information from the selected papers and the determination of which data is important for the research being carried out. In this sense, several MQs were defined:

- MQ1. Which are the main applications of PD&R in developing STEM Education?
- MQ2. What benefits does the development of STEAM Education derive from the application of PD&R and active methodologies?
- MQ3. How can the studies be classified according to the type of PD&R employed (physical devices and robots and the possible types)?
- MQ4. How can the studies be classified according to the collaborative methodology applied (ChBL, PBL, and PrBL)?
- MQ5. How can the studies be classified according to the educational context?
- MQ6. Which are the most relevant authors, sources (journals, conferences, books), and locations?

By answering these questions, it will be possible to gather the information that will allow us to answer the RQs and, therefore, to achieve the SMP goal.

2.2 | Conducting stage

The conducting stage consists of the application of the outcomes of the planning stage steps. In Figure 1 it is possible to see the different steps of this stage. A part of these is grouped in the PRISMA (Preferred Reporting Items for Systematic Reviews and Meta-Analysis) Model [80]. The steps included in the conducting stage are:

- *Identification.* This step includes the search process, requiring the application of the search string in the different sources defined in the planning, and the import of the studies. During this step, duplicate studies were removed.
- *Screening.* This step involves the application of the inclusion and exclusion criteria and the selection of the papers that can be interesting for research.

- *Eligibility.* This step includes the review of the reference list of the papers to find others that could be also included in the analysis as well as application of the quality assessment checklist.
- *Content.* Definition of the concrete papers that will be analyzed in the next two steps.
- *Data extractions.* With the selected papers, we answer the questions defined in the data extraction step, which allows us to gather the relevant information for analysis.
- *Data analysis.* Presentation of the results and determination of how they can answer the RQs defined at the beginning.

These steps can be grouped to present the conducting results using a PRISMA flow diagram [80] as shown in Figure 2. This diagram shows that after the identification phase (which includes searching, importing, and removing

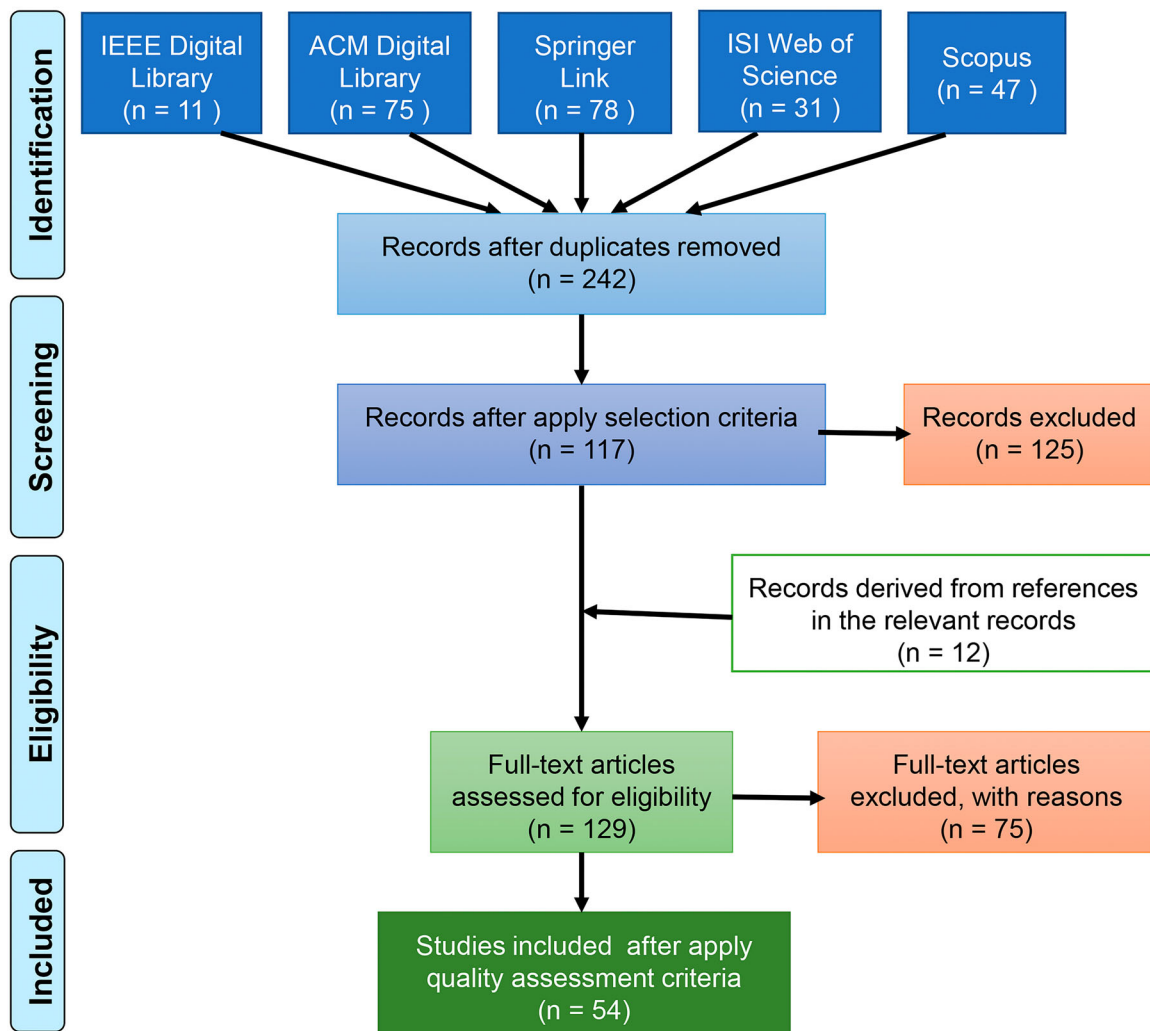


FIGURE 2 The Preferred Reporting Items for Systematic Reviews and Meta-Analysis (PRISMA) flow diagram adapted from Reference [80]

duplicates) we had 242 candidate studies. In the screening step, the inclusion and exclusion criteria were applied and the works reduced to 117. Later, in the eligibility phase, some new documents were only added after studying references [12] and the quality checklist was applied, finally resulting in 54 documents to be analyzed.

3 | DATA RESULTS AND DATA ANALYSIS

This section presents the results of the mapping from a quantitative perspective first. Later, the authors answer the questions defined in the data extraction form and analyze them.

3.1 | Quantitative distribution of the results

In any SMP, it is always interesting to have an idea of where the selected papers were found and the number

of publications per year. Figure 3 shows the distribution of selected and accepted papers per library, taking into account the inclusion, exclusion, and quality criteria. It is evident that the highest number of accepted contributions was found at Scopus and the highest percentage of accepted papers in IEEE. Other libraries, such as Springer Link and ACM have an important number of papers but only a low percentage of them were accepted. In this figure, the papers published by year is indicated. It is clear that from 2014 to 2019, there is a higher number of studies related to STEAM and PD&R, whereas from 2000 to 2014, there are only a few initiatives. Compared with the number of selected papers, there is a clear difference. There have also been more initiatives since 2000, although they are papers that were rejected as part of inclusion criteria.

3.2 | Data extraction and data analysis

In this section, the results of answering the data extraction form, that is, the MQs, are shown. It is necessary to take into account that this has been done only with the

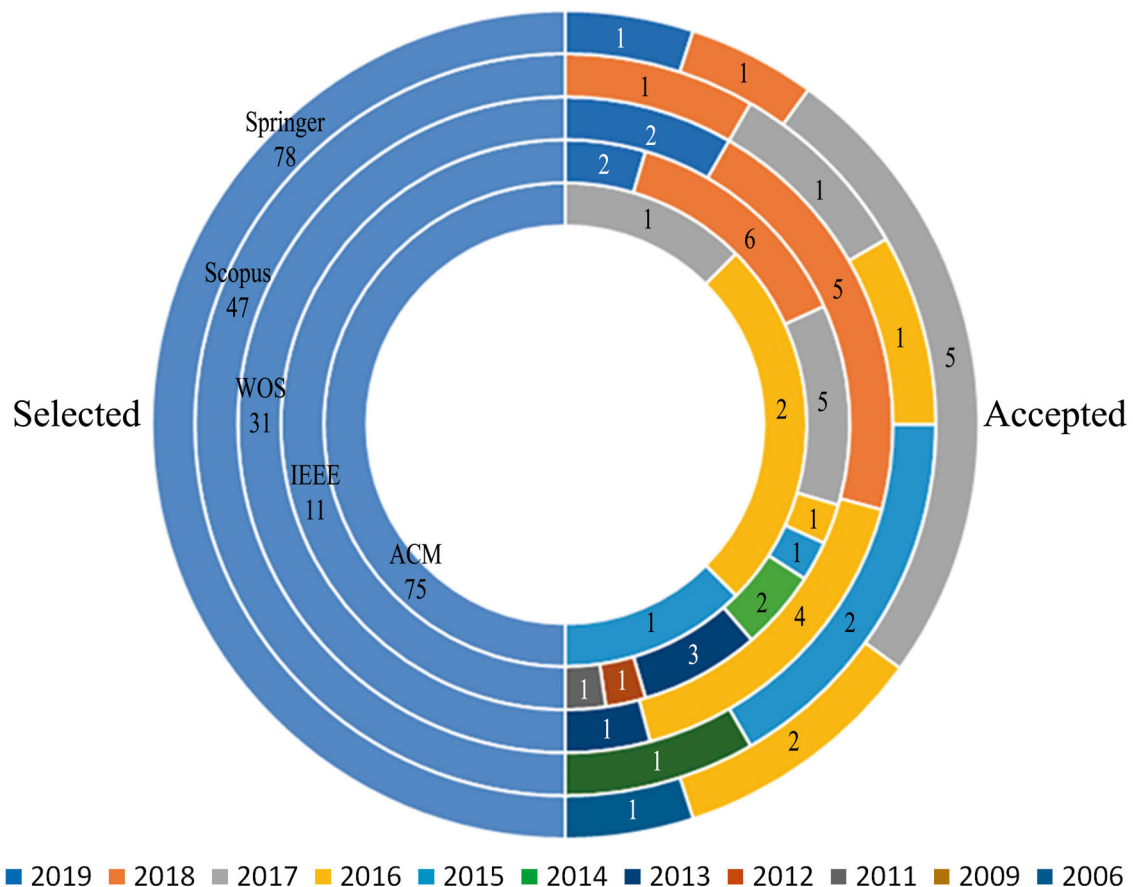


FIGURE 3 Distribution of publications per year and source

papers selected, after applying both the inclusion and exclusion criteria and the quality checklist.

3.2.1 | Which are the main applications of PD&R in developing STEAM Education?

To answer this question, the 54 selected papers, after applying the different criteria, were analyzed. In this case, the goals of each study were explored, and after checking all of them, it was possible to define four topics wherein the studies can be classified. These topics are:

- Topic 1: Studies focused on describing the application of PD&R in STEAM Education by employing active and collaborative methodologies such as PBL, PrBL, and ChBL.
- Topic 2: Studies devoted to describing the application of PD&R as a way to foster STEAM disciplines. These papers describe learning actions using collaborative methodologies to engage students with STEAM Education.
- Topic 3: Studies focused on the development of certain competencies such as Computational Thinking in the context of STEAM Education by employing methodologies such as PBL, PrBL, and ChBL.
- Topic 4: Studies that describe the application of different tools and techniques in STEAM Education but especially focused on the tools, such as PD&R, Evaluation tools, and so forth.

The classification of the studies by topics can be seen in Table 1. It should be noted that a paper can be included in more than one topic.

3.2.2 | What benefits does the development of STEAM Education derive from the application of PD&R and active methodologies?

Taking into account the selected studies, it is also interesting to understand which benefits are associated with the application of PD&R and active methodologies in STEAM are. We have analyzed and classified the studies according to the following topics:

- Benefits related to technology: In this sense, it is possible to consider benefits related to the application of some technology that is now actually affordable. Robotics and physical devices were years ago very expensive and complex and now can be employed as a powerful educational tool.
- Benefits related to methodologies: Other studies emphasize the effect of the application of active methodologies as a way to achieve valuable competencies such as problem-solving, teamwork, social or synthesis skills, and so forth. Most of these papers are focused on the application of such methodologies and the tools for doing it.
- Benefits related to specific competencies: Some research is more focused on the acquisition of very specific competencies, most of them related to STEAM, but also to other fields, which may lead to a lower dropout rate in scientific careers.
- Benefits related to new application contexts: Finally, some of the studies' authors understand that the main benefits of applying PD&R and active methodologies are the possibility of facilitating accessing STEAM Education in contexts with very specific features and even integrating collectives with special needs or those

TABLE 1 Studies classification by topic about the application of PD&R in STEAM

Topic	References	No. of papers
1: Studies focused on describing the application of PD&R in STEAM Education	[11,36,49,79,81,84,104,112]	8
2: Studies devoted to describing the application of PD&R as a way to foster STEAM disciplines	[11,12,20,30,35–37,44,48,52,57,61,64,65,70,73,74,76,78,103,104,107,111]	23
3: Studies focused on the development of certain competencies in the context of STEAM Education	[14,16,22,31,42,44,51,57,58,63,70,76,86,92,105,107,109]	17
4: Studies that describe the application of different tools and techniques in STEAM Education	[10,20,21,24,26,30,38,52,56,58,66,85–87,90,93,96,97,102]	19

Abbreviations: PD&R, physical devices and robotics; STEAM, science, technology, engineering, arts, and mathematics.

that traditionally are not interested in these types of disciplines.

This classification can be seen in Section A (Associated benefits) of Table 2. It should be noted that some papers discuss or deal with benefits that can be included in several categories.

3.2.3 | MQ3. How can the studies be classified depending on the type of PD&R employed?

After reviewing the selected papers, it was possible to classify them taking into account that they used PD&R. We have considered the use of robotics, the use of physical devices, and the use of both. Section B of Table 2 shows this classification. In these papers, the innovations related to the use of PD&R are also explored.

We should point out that the most popular devices were LEGO in different varieties such as LEGO NXT or LEGO MINDSTORM [35,38,66,70,86,103,107], and also, Arduino [10,42,85,92,93,96].

3.2.4 | MQ4. How can the studies be classified depending on the collaborative methodology applied?

Another interesting issue is the possibility to classify the methodology applied in each of the papers to look for usage patterns. To know which methodologies are more popular. Accordingly, the results are summarized in Section C of Table 2. It shows that PrBL is the most popular methodology and ChBL has not been applied in many studies. We should also point out that some studies have applied more than one methodology, such as References [11,107].

3.2.5 | MQ5. How can the studies be classified depending on the educational context?

One of the aims of this SMP is to know not only how PD&R is applied but also where it is applied. In this sense, it is interesting to know the educational level where the experiments described by the papers were carried out. Section D of Table 2 classifies the different papers into educational levels. We should point out that

TABLE 2 Distribution of selected studies for MQ2, MQ3, MQ4, and MQ5

Description	References	No. of studies
Section A—Associated benefits		
Technology	[21,35,51,52,61,76,78,79,81,85–87,93,97,109]	15
Methodologies	[10,11,20,21,24,26,30,31,37,38,44,48,49,51,61,64,65,73,74,76,81,85,87,90,93,104,109,111,112]	29
Competencies acquisition	[14,16,20,22,30,31,42,57,58,63,64,66,70,84,92,96,97,102]	18
Application Contexts	[12,36,42,48,58,70,79,81,84,85,87,103,105]	13
Section B—Tool employed		
Robotics	[12,16,24,31,35–37,48,49,52,56–58,61,63,70,73,76,78,81,84,86,87,93,97,103,109,112]	28
Physical devices	[10,14,21,38,42,44,51,65,74,90,92,96,102,107]	14
Both	[11,20,22,26,30,64,66,79,85,104,105,111]	12
Section C—Methodology		
PBL	[11,14,42,48,51,52,57,76,79,92,93,105,107,112]	14
PrBL	[10,11,22,24,30,31,35,36,38,44,56,58,61,64,65,73,74,78,81,84,85,87,90,102,103,107,109,111,112]	29
ChBL	[16,20,21,26,37,49,70,96,97]	9
Other	[12,66,86]	3
Section D—Educational level		
Elementary school	[57,81,86,93]	4
Middle school	[11,12,16,20,21,24,26,30,31,42,48,49,51,52,57,65,66,78,79,81,85,90,92,96,97,103,104,107,109,112]	30
High school	[10,44,52,58,61,79,81,84]	8
Higher education	[35,36,38,56,63,64,70,73,74,76]	10
Not specified	[14,22,37,87,102,105,111]	7

Abbreviations: ChBL, challenge-based learning; MP, mapping question; PBL, problem-based learning; PrBL, problem-based learning.

not all studies described specify this level, so they are included in the category “Not Specified.” From the table, it is possible to see more contributions at the Middle school level and fewer in Elementary school. There are some studies that explore their possible applications in the three K–12 levels [81] and some linked to more than one, such as [57] for Elementary and Middle school levels [52,79] or for Middle and High School. In the “Not Specified” category, we also have some special cases such as Taylor [105], who analyzes the students with special needs, or Burbaitè et al. [22], who consider the application in formal and nonformal learning environments. In this case, it is also possible to consider the impact of the use of PD&R based on the context.

We have also analyzed the number of students involved in the experiments—a total of 7,428 students—that involve a great number of students, especially competitions such as in References [26,37]. The maximum number of students involved is 2,500 and the minimum 5. The average is 189.4 and the standard deviation, 368.05. Some experiments were applied to different groups with different numbers of students and several times, so they have also a high number of individuals involved [12,16,37,107]. Figure 4 shows the studies distribution by students involved. It is evident that the most common activities involved less than 50 students.

3.2.6 | Which are the most relevant authors, sources (journals, conferences, books), and locations?

The last MQ explores the authors, sources, and locations of the papers.

Regarding the first issue, we have a total of 210 authors. The most relevant ones (the authors with more

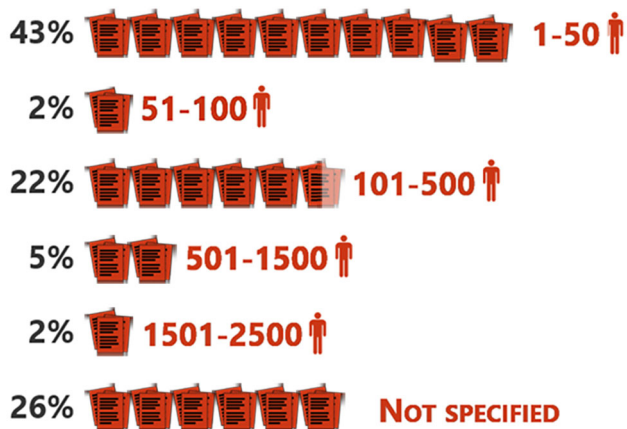


FIGURE 4 Percentage of studies per students involved in them

than one paper in the selected group) were Eguchi [36,37] and Gomoll et al. [48,49]. On the basis of the top 10 authors by Scopus and Google Scholar H-index, the results can be found in Table 3. The average Scopus H-index factor of the authors is 2.57 and the Scholar H-index 14.46.

Regarding the papers' sources and considering the type of publications, we found that 29 contributions were published in conference proceedings, 23 in journals, and two as book chapters. This means that 4% of the publications were book chapters, 42% journal papers, and 54% of studies included in proceedings. Surveying the different sources, the most relevant journal is the *Journal of Science Education and Technology* with three contributions. Taking into account the conferences, the most relevant ones are the ASEE (American Society for Engineering Education) Annual Conference and Exposition with four papers, the IEEE Integrated STEM Conference with three papers, and others with two papers, such as *Frontiers in Education*, the ACM technical symposium on Computer Science Education and the Annual Conference on Creativity and Fabrication in Education.

Regarding the authors' location, the distribution is shown in Figure 5. It should be noted that the researchers that work in this field are mostly from the United States of America, followed by Spain.

TABLE 3 Top 10 authors attending to the Scopus H-Index and Scholar H-Index

Author	Scopus H-Index	Scholar H-Index
Maurizio Porfiri	49	55
David L. Blustein	41	60
Cindy E. Hmelo-Silver	31	–
Illah Nourbakhsh	27	33
Robert Shin	27	–
Sertac Karaman	26	34
Yasmin B. Kafai	26	54
Manuel Castro	25	36
George Hademenos	23	–
Jane Connor	22	–
Muhammad Arshad	–	85
Chi-Cheng Chang	–	46
Alexander Repenning	–	36
Brian Magerko	–	28

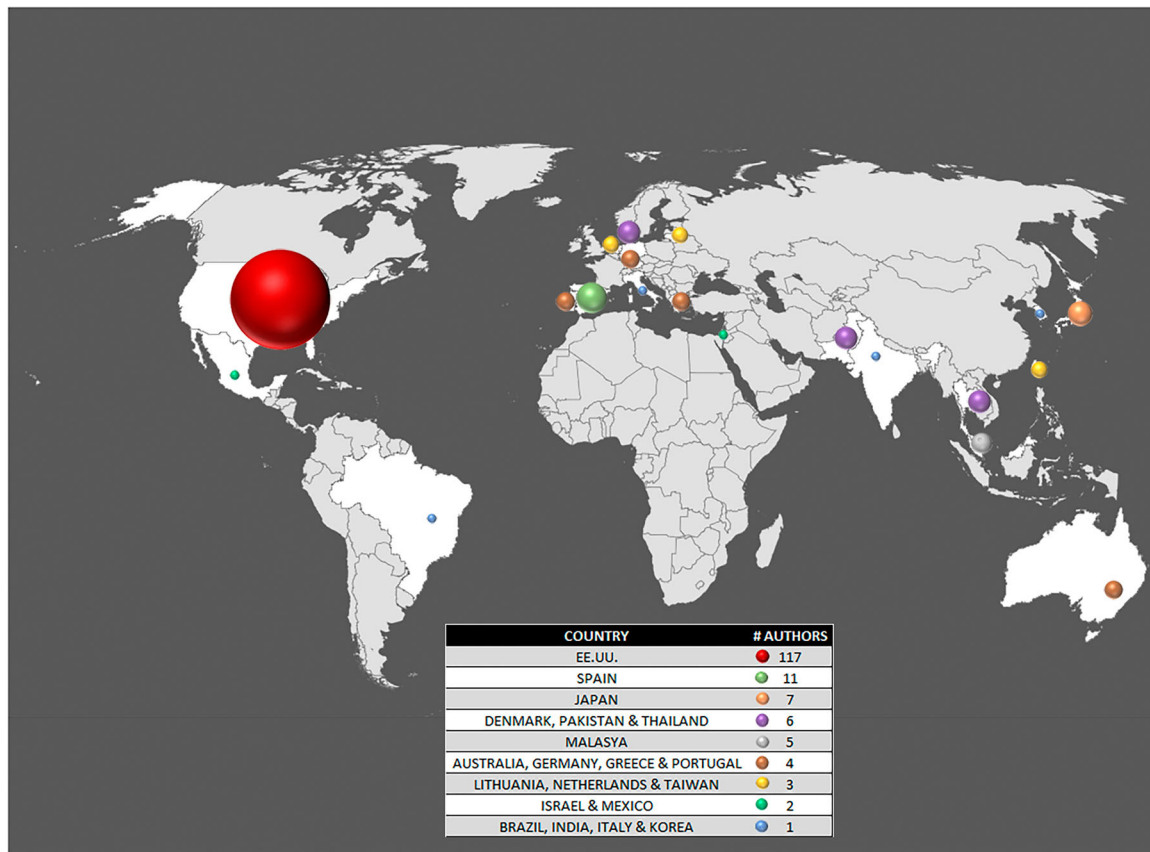


FIGURE 5 Distribution of authors by country of origin

4 | RESEARCH OUTCOMES AND DISCUSSION

Once the data extracted has been presented as results which answer the MQs, it is necessary to see how these questions facilitate answering the RQs of the SMP. The idea is to answer the RQs with the findings of the MQs, and in this way, accomplish with the mapping objectives. Figure 6 shows a conceptual map that summarizes how the MQs answer the RQs.

The different MQs will contribute to the definition of the general context of the research. Some of them will answer the RQs that define the academic context of the issue explored and that are necessary to achieve the research goals. MQ1 answers RQ1 related to the main approaches used, MQ2 answers RQ2 related to the benefits associated with the application of PD&R and active methodologies. Furthermore, MQ3, MQ4, and MQ5 answer RQ3 related to emerging tendencies in the explored field. MQ6, related to authors and publications support, as well as the other MQs, support the context of the research.

4.1 | RQ1. What are the main approaches used in STEAM Education which apply robotics and active methodologies?

The application of Robotics and active methodologies in STEAM Education is not new, this is shown by the different initiatives found during the mapping; in fact, we found 242 studies in the identification phase, including both those focused on STEAM Education and/or those that apply active or collaborative methodologies. Several of them were rejected after applying the inclusion and exclusion criteria and the quality checklist. This leaves 54 papers with different initiatives that address the mapping aims. By studying the studies' goals to ask MQ1, it was possible to classify them in different categories and with them answer RQ1.

In Figure 7, we have a comparison of a tag cloud with the titles of the papers and the categories that we defined, based on their goals. It is possible to see that although there is not a direct link, most of the main terms are related to the categories.

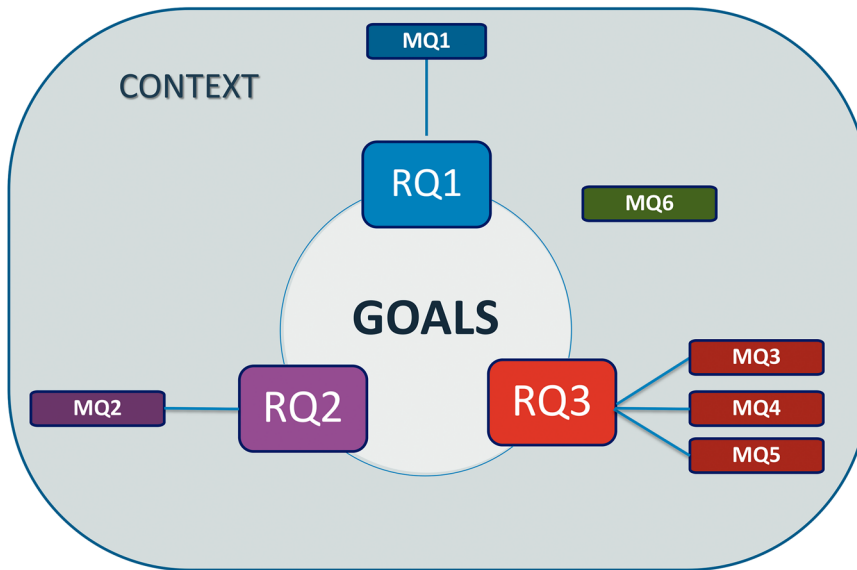


FIGURE 6 Conceptual map describing the relationship between mapping questions (MQs) and research questions (RQs)

Robotics is related to Topic 4 (Tool and Techniques). STEM, Engineering, Programming, and Design, are related to Topic 2, the idea of fostering the STEAM disciplines; school, students, or K12, may be related to the application of PD&R in a different context, that is Topic 1; and terms such as Project, Collaborative, Problem, or Learning are terms related to the methodologies employed (such as PBL, PrBL, or ChBL). This means that it is possible to see a relationship between the titles of the studies and the categories in which we divided them to answer RQ1 based on its goals.

Taking this into account we have classified the studies by the topics described for MQ1:

- *Topic 1.* The first topic considers studies focused on the application of PD&R in STEAM Education applying active methodologies. There are only eight studies describing such an application, which suggest that although both elements are employed in STEAM Education, their combination is not so common. In addition, it should be noted that these studies were published from 2015 to 2018, so they are very recent, which shows that this type of approach is quite new.

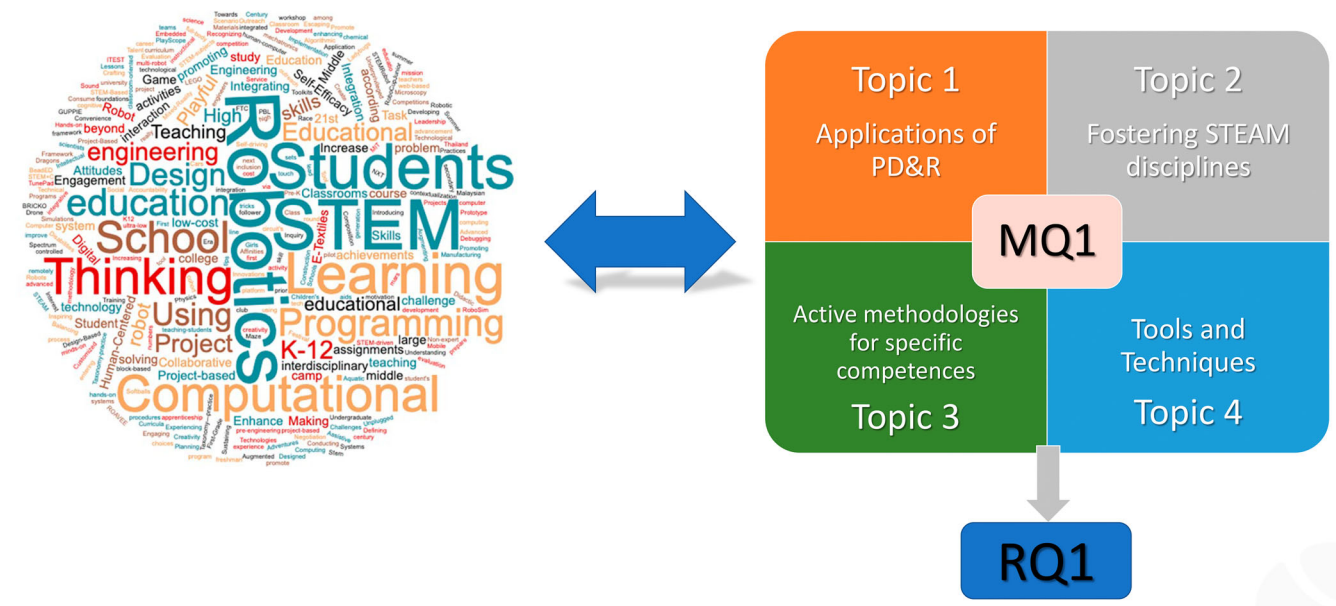


FIGURE 7 Comparison between a titles tag cloud and goal-based approaches categories

- *Topic 2.* The second topic explores an interesting issue to be taken into account, the students' engagement with STEAM Education. This can be motivated by the use of ICT and also with PD&R, as shown by the association of 23 studies in MQ1, a fact that agrees with some studies such as References [7,72], that show the positive effects of using the technology [107].
- *Topic 3.* The third topic is related to competence acquisition, and is, therefore, more focused on the results of the application of PD&R and the methodologies in the context of STEAM Education. As commented on in the Introduction, STEAM Education is linked to the development of certain competencies in accordance with the demands of the digital society [45,94] and the use of PD&R increases the students' motivation, which can facilitate competency development [98]. The studies included in this topic are especially devoted to describing these competency acquisitions and not so much the application of PD&R or the STEAM Education approach.
- *Topic 4.* The last topic is more focused on the tools or methodologies applied than on the experiment or the results obtained. There is a large number of studies classified under this topic (19 papers), which shows that teachers and researchers share their expertise with their peers to facilitate STEAM integration, which requires looking for resources, methodologies, tools, and so forth [101,108].

As the research outcome that answers this question, we can consider that after the analysis of these studies, their classification and the educational context where they developed; it is possible to see different approaches of the application of PD&R. The descriptions of each of these perspectives are equally valuable for the research. With the results obtained for Topic 1, we can conclude that there are few studies that describe the application of PD&R and active methodologies for STEAM Education, which means that there is an interesting gap to be filled by new studies. The second topic shows that in some cases the method followed is not so important but rather the competencies achieved, so the methodology or tools used should ensure the proper outcomes. Regarding the third topic, it is possible to see that STEAM Education requires students' engagement, and elements such as PD&R could be very useful to this end. Finally, as Topic 4 shows, with what others have used and done, it is possible to consider different approaches and the advantages and inconvenience in applying one or the other. Each of these topics could lead to define a question to be answered in a future systematic literature review.

4.2 | RQ2. Which are the potential benefits of changing the way of conducting STEAM Education approaches that apply PD&R and active methodologies over the years?

If we talk about benefits related to the application of different tools and methods and STEAM Education, initially we must consider the benefits related to STEAM Education and why this trend emerged. STEAM can be seen as the evolution of STEM, and that is not a new concept. The first appearance of the term in 1990 as SMET was in the United States [82] which has been promoting it as a key issue in their educational system [94]. However, nowadays it has become a global priority [25]. For example, in Europe, funds have been devoted to initiatives oriented toward developing effective and attractive STEAM curricular and teaching methods such as ENQUIRE [75], Mind the Gap [53], PRIMAS [1]. These are approaches that aim to improve teacher education and professional development related to STEM such as STELLA [99] or GRID [77]; and other related attempts toward making young people interested in STEAM, such as STEM Alliance [100] and ingenious [59]. This means that, for educational institutions, STEAM Education has become essential, a fact mainly motivated by this type of education fostering the acquisition of, what we have described previously as 21st-century skills. So, most of the studies related to STEAM Education will have as a final aim that students develop such skills. However, during the SMP, it was possible to see that these are not the only benefits. This is explored by MQ2, which classified the benefits into different categories. However, it would be interesting to go deeper into this issue.

The main benefits observed can be described as follows:

- **Benefits related to technology:** Technology is continuously evolving in capacity, functionality, and so on, but also becoming more affordable. This allows equipment such as robots or physical devices to be integrated in an easier way in schools, and the advantages, as reported by several authors in the literature, provided by educational robotics can be leveraged [15,43]. Several of the papers explored in this mapping focused their research goals on the application of such a technology to achieve STEAM-related competencies [51,61,78], and especially on the possibility of using such technology, to achieve different competencies in preuniversity education [61,78,79]. They provide different examples of the benefits of low-cost robots [35,52,76,86,87,93,97], open hardware and software

proposals, or other kinds of manufacturing technologies [51].

- Benefits related to the methodological application: In the context of STEAM Education, active methodologies have been very popular because of their benefits [46]. In these methodologies, the traditional students' and teachers' roles change. The students assume a central role and take responsibility for their own learning, are supposed to look for and define solutions for different problems and collaborate with peers [41], while the teacher acts as a guide in the learning process [17]. This facilitates the acquisition of competencies such as autonomous work, responsibility, leadership, project management, teamwork, and the ability to synthesize [91]. Taking this into account, it is normal that several of the papers were more focused on the methodology followed and the associated benefits. Of these, several are related to PrBL [24,30,31,38,61,64,65,81,85,87,90,109,111,112], PBL [49,73,76,93], or ChBL [20,21,26]. In addition, some studies state that this type of methodological innovation requires training [11,74,104], and specific curricular changes [49,73,76,93], so it is necessary to adapt the learning pathways judging from an institutional point of view.
- Benefits related to the acquisition of specific competencies: As this study explores STEAM Education, it is normal to find studies related to the competencies commonly associated with STEAM Education, such as computational thinking, problem-solving, critical thinking, programming, computing and design skills, interdisciplinary skills, computers science autonomy, autonomous learning, negotiating skills, and social abilities. Some examples of this type of papers are References [14,16,20,22,30,31,42,57,58,63,64,66,70,84,92,96,97,102]
- Benefits related to access to new contexts: One of the most relevant benefits of the changes in technology and methodology in STEAM Education is the possibility of carrying out activities in contexts, which previously were really complex. As commented above, the reduction in the price of robotics and physical devices has facilitated the application of such technology in preuniversity education, with different advantages related to an early acquisition of STEAM-related competencies [2,50], examples of papers that describe this as one of the main goals are References [12,42,58,79,81,103]. In addition, price reduction and the advantages related to the application of PD&R in STEAM Education also facilitate access to STEAM Education in new contexts such as in developing countries such as India [85]. Moreover, this also makes possible access to STEAM Education for minority students [84] and inclusion for several collectives

[36,48,87] and for disabled students [105]. Last, but not least, beyond the context or the inclusion of students, PD&R in STEAM Education increases the engagement of collectives traditionally not so attracted with STEAM careers, such as female students [36,70].

We should also mention that through analysis of the year of publication of each paper regarding the benefits, we can see how publications have evolved with time. This is shown in Figure 8. It is necessary to take into account that, as mentioned in Section 3.1, most of the selected studies were published after 2014.

The papers related to technology affordability were mostly published between 2017 and 2018 and related to the fact that hardware technology reduction has been very important in the last few years. Those studies with benefits related to methodological impact are more distributed in time although with an important peak in 2017. The distribution over time could be normal due to the fact that these methodologies are not new; however, active methodologies have been widely adopted in the last years due to the benefits associated with them and the technology available, which makes their application easier in the context of STEAM Education. The studies related to competencies acquisition are well distributed among the years, as STEAM Education associated competencies are well-known and there is an important effort to foster them through different initiatives and is an important investment, as commented above. Finally, the integration in preuniversity education is not so new, although most of the selected studies were published in the last 5 years, whereas the idea of integrating collectives or engaging some of them is more recent.

Following this analysis, it is clear that there are several categories of goals in the studies related to PD&R and STEAM Education, some more focused on methodologies and others on STEAM but also some more linked to the technology and to the possible application contexts.

4.3 | RQ3. Which are the emerging tendencies in the application of PD&R and active methodologies in STEAM Education?

The emerging tendencies in the application of PD&R and active methodologies in STEAM Education are related to the previous questions and have been answered by MQ3 and MQ4. We will consider again the tendency in issues related to technology, methodology, and educational context.

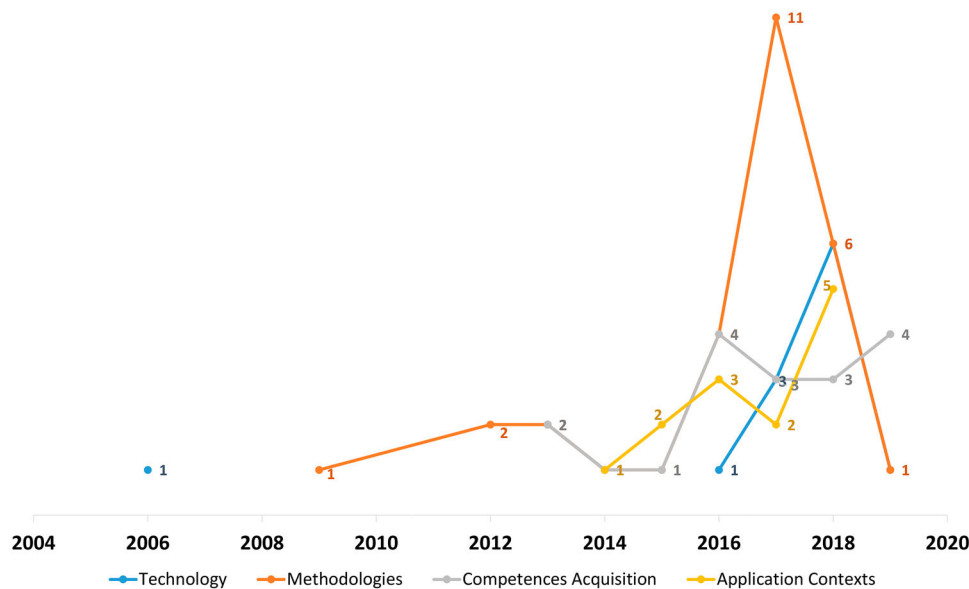


FIGURE 8 Distribution of studies publication classified by benefits with the time (calculated from the mapping data set [28])

From a technological perspective, the emerging tendency can be the application of robots in STEAM Education approaches mainly because (a) robots are really affordable now, some of which are low-cost with multiple functionalities, and (b) they are very popular. Companies such as LEGO Robots have promoted competitions related to STEAM [67] and have kits for STEAM Education such as LEGO MINDSTORM [68] or LEGO WEDO [69]. Regarding physical devices, the most popular one is Arduino, which also has an educational section with different kits [6]. The tendency in the research field is to apply certain types of devices and new active methodologies.

From a methodological point of view, the experiments carried out are farther from the trendy option that could be the newest, namely, ChBL. As previously commented, we have considered three of the most popular collaborative methodologies: ChBL, PrBL, and PBL. However, we did not limit the search to them. After applying the different selection criteria and using the data extraction form, we have classified the studies by methodology as shown in MQ4 and Table 2. In this table, the most popular methodologies are PBL and PrBL. The probable reason for this is that ChBL is quite recent. The ChBL foundations were described in the Apple ACOT reports about learning in the 21st century and the educational system in America in 2008 [4] and defined in 2009 [5], only 11 years ago. However, the other methodologies are older; PBL was firstly defined in 1980 [13] and PrBL in 1918 [62]. This means that PBL and PrBL are more popular and are more defined than ChBL so that these methodologies have more applications among the

selected studies in the SMP. In any case, PrBL has evolved and now is very close to what we understand as ChBL, “a project may be included in Challenge-Based Learning, and a Challenge may be part of a Project-Based Learning Experience” [83].

Another interesting issue to explore, and which can affect the type of approaches is the educational context and where it is carried out as commented in RQ3. MQ5 provides us information about this. After exploring the educational level where experiments were carried out, it is possible to point out that there is a tendency for development in Middle School, the rest of the categories have more or less the same number of applications, except for Elementary school students, where the number was only four. This does not meet the recommendation of different studies that say that STEAM Education should begin in Elementary school because it has a positive effect in perceptions and dispositions [8,23,32]; but agrees with different studies [34,39] that consider that it has not been applied at this level for several reasons: the nature of Elementary school teaching [55]; the need to provide secondary teachers with knowledge and resources about STEAM [106]; the maturity level of students and the contents to include in the curricula [110]. Given these facts, the next higher school level after Elementary school would be the Middle school, as it is normal to have more initiatives at this level. It should be pointed out that STEAM Education is not necessarily only applicable in formal environments [22] and that it also could be useful for students with special needs [105].

Another factor to take into account the context is the number of students involved. In this case, there are

different approaches in the SMP, especially taking into account the standard deviation from the average number. This is because some initiatives are only focused on their own experiments in a single class, for example, References [78,84]; others apply the same approach in several schools, for example, References [30,81]; and others show the results in competitions, for example, References [26,37]. Given this situation, the number of students involved in the different applications is probably not so relevant, and there is not a clear tendency regarding the type of initiative. What can be more interesting is that in 54 studies, we have a total of 7,428 students involved, which means that STEAM Education and the application of PD&R are having a high impact on our current educational contexts.

This means that the tendencies are mostly related to the technology evolution and the educational context in which the activity was carried out.

5 | CONCLUSIONS

STEAM Education is essential for our current educational systems as specified in several reports and shown by the quantity of funds and efforts devoted to this. However, the development of STEAM Education and especially the integration into current educational curricula is not easy. It is necessary to engage young students in these disciplines, which is sometimes difficult because STEAM cannot be understood as a subject or a set of them, but as disciplines that need to be included in daily education.

The present paper has aimed to describe the landscape in the field both from a general and from an academic perspective so that it can later be useful for initiatives such as RoboSTEAM Project to facilitate the development of computational thinking and the integration of STEAM by using attractive tools for students such as robotics and physical devices and applying collaborative methodologies such as ChBL.

From our experience, several conclusions can be drawn:

- The application of PD&R in STEAM Education is quite common and increases students' motivation, which may improve their learning and also increase their engagement with STEAM disciplines. There are different applications in this sense, but given the benefits provided by them, it is necessary to promote these types of activities.
- Active methodologies are helpful for STEAM integration because they are quite flexible. The most common one is PrBL but as ChBL is very similar but newer and focused

on 21st-century competence acquisition, it will become the norm that, in the near future, the number of applications of this methodology would be increased.

- The tendency in this field is to apply low-cost robots in STEAM Education and foster STEAM-related competencies through competitions. As this is carried out by using active and collaborative methodologies, the integration into educational plans is easy. However, this should be a continuous process, with special emphasis on the early stages of the educational process. To facilitate this, it is necessary to provide students and teachers with the tools, resources, and support because Elementary education is a very particular context that has its own features.
- STEAM Education happens not only in formal contexts but beyond educational institutions, this can be explored by looking at new application areas.

Taking these findings into account, the RoboSTEAM project development can clearly be an advance in the field of STEAM Education. It explores the application of tools and methodologies, the production of outcomes that can be used by teachers and students at different school levels, and it also provides support to STEAM Education practitioners.

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DATA AVAILABILITY STATEMENT

The data that support the findings of this study are openly available in Zenodo at <https://doi.org/10.5281/zenodo.4048086> and in Reference [28].

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